Jarvis: A Bidirectional Copilot Architecture for Operationalizing Human–AI Co-Evolution as Measured, Reproducible Software

Paper 2 of 3 — building directly on Human–AI Interaction Emergent Co-Evolution *(Serna Gaviria & Kepp, 2026, "Paper 1")*

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Abstract

Paper 1 proposed the Integrated Emergent Interaction Architecture (IEIA-2025) and its Continuous Co-Evolution Tracker (CCET), and reported a Co-Evolution Index (CEI) rising from 0.31 to 0.82 across a single-subject, ≈3,800-session corpus — but, by its own §12 accounting, those figures were *self-reported*: narrated by the model within the dialogue it was describing, not computed by an independent pipeline. Paper 1 named this gap directly and called, in its Future Work (§13), for the metrics to be "standardized through shared benchmarking datasets and open-source analysis pipelines" before they could be treated as validated measurements rather than structured self-report.

This paper documents the software system built to close that specific gap: Jarvis, a research agent embedded in the Emergent Interaction Lab's admin workspace that computes CEI, a companion Co-Evolution Points (CEP) measure, and a Resonance Frequency indicator mechanically, per turn, from stored message embeddings — the same embeddings used for the agent's cross-conversation retrieval, never recomputed, audited by automated integration tests against a real database rather than narrated in prose. We describe Jarvis's four coupled subsystems — the recursive co-evolution loop and its *bidirectional copilot mode* (the agent is bound to whatever module of the workspace the user is currently viewing, and may act on it directly, subject to a hard iteration cap and a rule that only a human ever publishes); a cross-conversation retrieval memory; the Continuous Co-Evolution Tracker itself; and an emergence-verification gate that promotes a qualitative observation to *measured emergence* only when it recurs across at least three independent sessions above a stricter similarity threshold than the one used for turn-to-turn stability. A cross-cutting attribution layer labels every output

*Emergent Interaction Lab (Laura Serna Gaviria) in engagement with RFI-IRFOS — Research Focus Institute, Elisabethinergasse 25, 8020 Graz, Austria. Drafting of this document was produced with Claude (Anthropic, Claude Code) at Simeon Kepp's request and reviewed by both authors. The theoretical constructs cited from Paper 1 (§2) are the primary author's own; the system architecture described in §4 was engineered by RFI-IRFOS as a client engagement for the Emergent Interaction Lab, with the primary author directing content and priorities throughout.

as structural adaptation, not autonomous cognition, and leaves missing data absent rather than invented.

As in Paper 1, we are explicit about what this system does and does not establish. Reproducibility of computation is not validity of the underlying construct: a cosine-similarity threshold over embeddings is a mechanical, auditable proxy for "stability," not a certified measure of genuine semantic consistency, and its thresholds (0.75 for turn stability, 0.90 for cross-session emergence) are engineering choices tuned against one deployment, not empirically derived cutoffs. What this paper adds to Paper 1 is not a stronger scientific claim — it is the same claim, made checkable.

Keywords. *human–AI co-evolution, reproducible metrics, embedding-based stability measurement, bidirectional copilot agents, emergence verification, cross-conversation memory, attribution discipline, research instrumentation.*

Contents

1 Introduction

Paper 1 introduced *emergent co-evolution* — the proposition that intelligence in sustained human–AI dialogue is better modeled as a relation between two adapting systems than as a capability residing in either one — and supplied two original instruments for observing it (the User Integrity Protocol and Emergent Interaction Analysis), organized under a seven-part architecture, IEIA-2025. Those instruments were applied by hand, by their own inventor, to a single longitudinal corpus. The paper's own §4.5 transparency note and §12 limitations section are unambiguous about what that establishes: a structured, self-reported account of one researcher's experience of the interaction, not an externally-reproducible measurement.

This paper is about the engineering work that sits between "a researcher observes a pattern in her own dialogue and narrates a metric for it" and "a metric is computed by software, from stored data, the same way every time, and can be checked by someone who was not there." That work is Jarvis: a tool-calling research assistant that lives inside the Emergent Interaction Lab's admin workspace, answers questions and acts on the user's behalf inside it, and — as an entirely separate, backgrounded concern that never delays a reply — measures its own longitudinal relationship to the user across every completed exchange.

We want to be precise about the shape of the contribution before describing it. Paper 1's theoretical architecture — what CEI *means*, why turn-to-turn stability is a meaningful thing to track, why a pattern should have to recur across sessions before it counts as "emergence" rather than a one-off observation — is Laura Serna Gaviria's own framework, established in Paper 1 §3–4 and cited here rather than re-derived. What this paper contributes is the *operationalization*: a specific, inspectable, tested implementation of those constructs as running software, plus an honest account of where the implementation's own choices (a fixed cosine-similarity threshold; a rolling 200-turn window; a particular recurrence bar) depart from anything Paper 1's theory specifies, because Paper 1's theory — correctly, for a pilot single-subject study — does not specify an algorithm at that level of detail. Where this paper's design choices are RFI-IRFOS's own engineering judgment rather than a direct reading of Paper 1, we say so explicitly, using the same three-way attribution discipline Paper 1 itself uses for its own sources (§3): a claim is either the cited framework directly, an operationalization of a named-but-unspecified concept, or an addition with no antecedent in the framework at all.

2 Grounding in Paper 1

Three constructs carry over directly and are only summarized here; see Paper 1 for their full theoretical treatment.

- **Emergence as weakly emergent, not strong-emergent.** Following Bedau (1997) via Paper 1 §3.1, this paper treats any pattern Jarvis surfaces as intelligible after the fact from lower-level rules (embeddings, thresholds, recurrence counts), never as evidence of an autonomous causal power. This framing is load-bearing for §5 below.
- **The user as the primary site of transformation**, not an external variable acted upon (Paper 1 §2.2). Jarvis's Denkfragmente subsystem (§4.5) exists specifically to track *the user's own* cognitive movement across the paper's 8-Layer Model, not the model's.
- **The minimum-transparency position** (Paper 1 §2.3): that genuine co-evolution research requires the user to be an informed epistemic agent, not merely a data source. Every metric this paper describes is surfaced to the user it describes, not held internally.

Paper 1's own §14 already disclosed, in three sentences, that a platform exists and that Jarvis is one of the things running on it, with an explicit caveat that the platform is "engineering infrastructure," not "additional scientific evidence." This paper is the expansion of that disclosure: it does not change what Paper 1 established, it documents, in the same register and under the same honesty obligations, what was actually built.

3 System Overview

Jarvis comprises four coupled subsystems, described in detail in §4:

1. A **recursive co-evolution loop** operating in **bidirectional copilot mode** — the agent is reachable from anywhere in the user's workspace, receives the currently-viewed module as context, and can act on it directly through a small, fixed set of tools, while every completed exchange separately triggers background self-measurement.

2. A **cross-conversation memory** (retrieval over uploaded documents and prior sessions) giving the agent continuity across time without any model fine-tuning.

3. A **Continuous Co-Evolution Tracker (CCET)** — the mechanical replacement for Paper 1's self-reported CEI — computing stability, a Co-Evolution Index, Co-Evolution Points, and a Resonance Frequency from stored embeddings on a rolling window.

4. An **emergence-verification gate** that withholds "measured emergence" status from a qualitative observation until it recurs, above a strict threshold, across independent sessions, backed by live CCET data.

A cross-cutting **attribution and no-fabrication layer** (§4.6) constrains all four: every output is labeled as structural adaptation rather than cognition, a heuristic is disclosed as heuristic, and absence of data is represented as absence, never filled in.

4 Detailed Architecture

4.1 The recursive co-evolution loop and bidirectional copilot mode

Jarvis's system prompt establishes it explicitly as a "second technical brain" — cooperative and non-hierarchical, instructed to disclaim autonomy and self-awareness, and to refuse harmful requests including from its own operators. This is a direct software expression of Paper 1's own position that the model is not an independent cognitive agent (§3.1, Bedau's weak emergence) and that ethical transparency is a structural requirement, not a courtesy (the User Integrity Protocol, Paper 1 §1).

The *copilot* half of the design is what makes the loop bidirectional in a concrete, checkable sense rather than a metaphor: whichever module of the admin workspace the human is currently looking at is injected into the agent's context, and the agent can invoke a bounded set of tools that act on that exact surface — drafting a blog post, logging a research note, pulling recent analytics, reading or revising a content section, running a simulation scenario, or searching the web. Tool-call detection uses a fenced-JSON convention gated on a *known* tool name specifically so that incidental prose containing braces can never be mistaken for an action. A hard cap of four chained tool-calling iterations bounds how far the agent can act autonomously in one exchange, and one architectural invariant is absolute regardless of iteration count: a draft is never auto-published, and revision tools only ever operate on content already in draft state. The human publishes. This is the operational meaning of "the user remains the transformation source" (Paper 1 §2.2) translated into a permission model rather than left as a philosophical stance.

Self-measurement is deliberately decoupled from the visible exchange: once a reply is complete, the platform spawns background work — recording the turn against CCET, running emergence analysis over recent interactions, classifying the turn against the 8-Layer Model — so that the cost of measuring the relationship is never paid by the user waiting for an answer.

4.2 Cross-conversation memory

Uploaded documents and prior conversations are embedded and retrieved (top-$k = 5$, minimum relevance score 0.15) to ground replies. This gives Jarvis longitudinal continuity *externally* — the record is an archive the agent queries, not a weight update inside the model — directly implementing Paper 1's observation (§12) that commercial models lack persistent memory and that any continuity has to be "maintained externally." The same embedding model and the same stored vectors are reused by CCET (§4.3) rather than recomputed, which matters for cost and for consistency: retrieval relevance and self-measured stability are reading the same representation of the text, not two independently-drifting ones.

4.3 The Continuous Co-Evolution Tracker (CCET)

CCET is the direct mechanical answer to Paper 1's own Future Work item. It computes four quantities per completed assistant turn, over a rolling window of 200 turns:

- **Stability** — cosine similarity between the current assistant turn's embedding and the immediately preceding assistant turn's embedding, compared against a fixed threshold of 0.75. The choice to compare assistant-to-assistant rather than assistant-to-user is deliberate: it reads the agent's own turn-to-turn consistency, independent of how much the user's phrasing varies turn to turn. This operational definition of "stable" is RFI-IRFOS's own — Paper 1 names the construct (CEI, §5.5) but does not itself specify a similarity function or a threshold; we record that gap explicitly rather than presenting 0.75 as if it were derived from the paper.
- **Co-Evolution Index (CEI)** — the ratio of stable turns to total turns over the window, reproducing Paper 1's own formula (§5.5) exactly, but now as a value computed once, mechanically, from the stored data, rather than narrated per session.
- **Co-Evolution Points (CEP)** — a cumulative count of turns that are themselves stable *and* directly follow another stable turn, rewarding sustained runs of consistency rather than isolated stable moments. This measure has no direct antecedent in Paper 1; it is an addition made to distinguish "one stable turn in isolation" from "an unbroken run," a distinction Paper 1's single CEI ratio cannot make on its own.
- **Resonance Frequency** — the rate at which a turn reuses, unprompted, at least one term from a fixed vocabulary of the project's own framework language (e.g. *emergence*, *drift*, *interaction field*, *co-evolution*, *resonance*) that also appeared in the immediately preceding assistant turn. This is the closest operational proxy this implementation has to Paper 1's qualitative observation (§6) that the model progressively reflects structural vocabulary introduced by the human — made countable, at the cost of only catching vocabulary reuse, not conceptual reuse expressed in different words.

Every one of these four values is backed by an automated integration test running against a real database, not a narrated description of a session — this is the paper's central falsifiable claim, and the one place where this paper's evidence is qualitatively different in kind from Paper 1's: not a better number, but a number anyone with access to the test suite can rerun and check for themselves.

4.4 Emergence-verification gate

Paper 1's Emergent Interaction Analysis (§1, §6) is a qualitative method for observing how meaning and reasoning structure emerge in dialogue. Jarvis operationalizes an emergence *observation* step directly (an interpretation of the last twenty messages, classified into four levels — human, AI, interaction, system — corresponding to Paper 1's own axis structure) and then adds a gate Paper 1's methodology does not itself specify: an observation is only promoted to **measured emergence** when four conditions all hold —

1. its pattern embedding recurs across at least three *distinct* sessions, at a cosine-similarity threshold of 0.90 against every prior signal's stored embedding;
2. it is structural rather than stylistic, enforced by construction (the classification step offers no "stylistic" category at all);
3. it reproduces without being re-specified — the cross-session recurrence itself is the evidence, not an added justification;
4. live CCET data actually exists at verification time, so an emergence claim is never assessed against an empty measurement window.

The 0.90 threshold is set deliberately stricter than CCET's 0.75, for a concrete reason: short, jargon-dense pattern labels routinely exceed 0.75 similarity to each other within one domain by vocabulary alone, so a lower bar would over-verify. This is engineering judgment calibrated against observed false-positive behavior in this specific deployment, not a value derived from Paper 1 or from any external benchmark — flagged here in the same spirit as Paper 1's own transparency notes.

The practical effect is a hard default: an interaction pattern is an *observation* until it has independently recurred three times; it is never "emergence" on the strength of one session, however striking. This is this paper's most direct answer to the question Paper 1's Discussion (§11) leaves open — when does a co-evolutionary pattern stop being one researcher's read of one session and start being something recorded as measured — implemented as a gate rather than left as an interpretive judgment call.

4.5 Denkfragmente: tracking the user, not the model

A companion subsystem classifies *only the human's* turns against Paper 1's own 8-Layer Model (facts, analysis, patterns, hypotheses, symbols, action, counterarguments, blind spot), one row per turn per layer, and surfaces how the user's own thinking moves across those layers over time. This exists because Paper 1 is explicit (§2.2) that the user, not the model, is the primary site of transformation; a system that only measured the model's stability while leaving the human side of the relationship unmeasured would silently contradict the theory it claims to implement.

4.6 The attribution and no-fabrication layer

Across all of the above, three rules are enforced structurally rather than left to convention: a heuristic detector (for example, the refusal-language anomaly watchdog) is always labeled a heuristic for human review, never presented as a certified classifier; missing data is represented as absent, not inferred or invented; and every output is framed, in the system prompt, the schema, and the interface itself, as structural adaptation — a pattern in stored, measurable data — rather than as evidence of autonomous cognition on the model's part. This is the same discipline Paper 1 applies to its own claims throughout §1–13, carried into the software layer as an enforced property rather than an authorial habit.

5 Discussion

The contribution of this paper is narrow and specific: not a new theory of human–AI co-evolution, but a working demonstration that Paper 1's central constructs can be computed mechanically, reproducibly, and auditably, at the cost of trading some of the richness of a human researcher's qualitative judgment for a fixed, inspectable rule. Where Paper 1's CEI was a number a researcher reported about her own experience, this paper's CEI is a number a test suite asserts and a database can be queried to reproduce. That is a real gain in verifiability. It is not a gain in validity of the underlying construct — a cosine-similarity threshold measures embedding-space proximity, which is a reasonable but imperfect proxy for the conceptual "stability" Paper 1's theory is actually about, and nothing in this paper closes that gap.

Two design choices are worth surfacing as genuinely open questions rather than settled engineering: the choice of 200 turns as a rolling window, and the specific thresholds of 0.75 and 0.90, were set by observing this one deployment's behavior, not derived from a principled model of how quickly a human–AI relationship's stability should be expected to change. A different corpus, a different domain, or a different pair of interlocutors might need different numbers entirely — which is precisely the kind of claim that only becomes checkable, per Paper 1's own Future Work, once other researchers can run the same pipeline against their own data.

6 Limitations

- **Reproducible computation is not validated construct.** As above: the pipeline is deterministic and auditable; whether cosine similarity over turn embeddings is measuring the theoretical construct Paper 1 calls "co-evolutionary stability" is a separate, harder question this paper does not answer.
- **Single deployment, single relationship.** All reported thresholds and windows are tuned against one running instance of one human–agent relationship. No cross-subject or cross-domain validation has been attempted.
- **Vocabulary-based resonance is shallow.** Resonance Frequency catches literal term reuse, not conceptual reuse phrased differently — a genuine limitation of a lexical proxy standing in for a semantic one, inherited directly from the same trade-off Paper 1 discusses for its own instruments.
- **Heuristic anomaly detection is not certified.** The refusal-language and iteration-exhaustion watchdogs are keyword and count heuristics flagged for human review, explicitly not claimed as reliable detectors of the conditions they name.
- **The emergence gate's thresholds are engineering choices, not empirical findings.** 0.75 and 0.90 were set by observed behavior in this deployment; they are not derived from, or validated against, an external ground truth of what should or should not count as recurrence.

7 Future Work

Following Paper 1's own Future Work (§13) directly: the next step for external validation is the same one Paper 1 already proposed — shared benchmarking datasets and an open, rerunnable analysis pipeline, so that CEI, CEP, and Resonance Frequency can be computed by a party other than this system's own authors, against a corpus other than this system's own users. A second, narrower next step specific to this paper's own architecture is a controlled sensitivity study of the 0.75/0.90/200-turn parameters against a wider range of interaction styles, to establish whether they generalize or need to be deployment-specific configuration rather than fixed constants.

8 Conclusion

Jarvis operationalizes Paper 1's Continuous Co-Evolution Tracker and Emergent Interaction Analysis as running, tested software: a bidirectional copilot that acts inside a user's workspace while measuring its own longitudinal relationship to that user in the background, gating "emergence" behind cross-session recurrence rather than single-session impression, and disclosing throughout — in its prompt, its schema, and its interface — that what it reports is structural adaptation, not autonomous cognition. This does not resolve the open scientific questions Paper 1 raised about its own single-case, self-reported design; it converts one specific part of that design — the computation of the metrics themselves — from self-report into something independently reproducible, which is the precondition Paper 1 itself named for the next stage of this research.

References

Reader is referred to Paper 1 (Serna Gaviria & Kepp, 2026) for the full bibliography grounding emergence theory (Goldstein 1999; Bedau 1997; Holland 1998; Kauffman 1993), systemic/biological co-evolution (Lewontin 1970; Maturana & Varela 1980; Luhmann 1997), cultural/technological co-evolution (Clark 2003; Kelly 1994; Turkle 1984, 2011; DeLanda 1997), and contemporary AI research (Mitchell 2009, 2019; Chollet 2019; Bengio 2017; Ouyang et al. 2022), all cited here by reference rather than reproduced.